

$^{47}\text{Sc} \beta^-$ decay **1986Re12,1974HeYW,1956Gr12**

Type	Author	History	Citation	Literature Cutoff Date
Full Evaluation	S. Ota and E. A. Mccutchan		NDS 203,1 (2025)	1-Apr-2025

Parent: ^{47}Sc : $E=0.0$; $J^\pi=7/2^-$; $T_{1/2}=3.3492$ d 6; $Q(\beta^-)=600.8$ 19; $\% \beta^-$ decay=100

1986Re12: ^{47}Sc activity from Institut National des Radioelements, Fleurus, Belgium. Measured E_γ , I_γ , $\gamma(t)$ using HPGe detectors and absolute activity using a $4\pi\beta$ - γ coincidence technique and liquid scintillators.

Others: **1974HeYW**, **1972GeZG**, **1956Gr12**, **1949Kr12**, **1953At22**, **1953Ch16**, **1953Co44**, **1953Du22**, **1953Ma64**, **1955Ly34**, **1955Ni15**, **1956Gr12**, **1956Li38**, **1959Po64**, **1963Ho17**, **1964Mi07**, **1967Ko01**, **1968Ba33**, **1968Me07**, **1969Ra16**, **1969Wo02**, **1980Mo26**.

 ^{47}Ti Levels

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>$T_{1/2}$[‡]</u>
0.0	$5/2^-$	stable
159.381 15	$7/2^-$	210 ps 6

[†] From E_γ .

[‡] From the Adopted Levels.

 β^- radiations

av $E\beta$: [Additional information 1](#).

<u>E(decay)</u>	<u>E(level)</u>	<u>$I\beta^-$[†]</u>	<u>Log ft</u>	<u>Comments</u>
(441.4 22)	159.381	68.6 4	5.297 7	av $E\beta=142.0$ 7 E(decay): others: 439.1 16 (1986Re12), 450 10 (1956Li38), 439 2 (1956Gr12), 430 5 (1955Ni15), 460 20 (1955Ly34), and 435 8 (1953Ch16). 280 (1953Ma64 is discrepant). $I\beta^-$: others: 67.4 14 (1986Re12 , β spectrometer measurement), 66 4 (1953Ch16), 28 (1953Ma64), 66 3 (1955Ly34), 64 (1955Ni15), 60 (1956Gr12), 74 (1956Li38), and 68.5 27 (1968Ba33).
(600.8 24)	0.0	31.4 4	6.113 7	av $E\beta=203.3$ 8 E(decay): others: 600.15 19 (1986Re12), 610 5 (1956Li38), 600 2 (1956Gr12), 596 10 (1955Ni15), 620 30 (1955Ly34), 622 5 (1953Ch16), and 640 39 (1953Co44). 490 (1953Ma64 is discrepant). $I\beta^-$: others: 32.6 14 (1986Re12 , β spectrometer measurement), 34 4 (1953Ch16), 72 (1953Ma64), 36 (1955Ni15), 40 (1956Gr12), 26 (1956Li38).

[†] Absolute intensity per 100 decays.

 $\gamma(^{47}\text{Ti})$

I_γ normalization: Absolute intensity measured by **1986Re12**. Value is a weighted average of 4 individual measurements 67.8 5 and 67.1 8 (HPGe), 68.7 4 (NaI), and 68.9 7 (Ge spectrometer) (**1986Re12**).

<u>E_γ</u>	<u>I_γ[@]</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[†]</u>	<u>δ[†]</u>	<u>$\alpha(\text{exp})$^{‡#}</u>	<u>Comments</u>
159.381 15	68.3 4	159.381	$7/2^-$	0.0	$5/2^-$	M1+E2	-0.099 9	0.0045 3	$\alpha(\text{K})_{\text{exp}}=0.00406$ 21; $\alpha(\text{L})_{\text{exp}}\approx 0.000406$ 21; $\alpha(\text{M})_{\text{exp}}\approx 5.19\times 10^{-5}$ 31; $\alpha(\text{N})_{\text{exp}}\approx 2.78\times 10^{-6}$ 17 $\alpha(\text{K})=0.00560$ 11; $\alpha(\text{L})=0.000512$ 10;

Continued on next page (footnotes at end of table)

$^{47}\text{Sc } \beta^- \text{ decay } \quad \text{1986Re12,1974HeYW,1956Gr12 (continued)}$ $\gamma(^{47}\text{Ti})$ (continued)

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>Comments</u>
		$\alpha(\text{M})=6.54 \times 10^{-5} \text{ } 13$; $\alpha(\text{N})=3.49 \times 10^{-6} \text{ } 7$ E_γ : from 1972GeZG and 1974HeYW . Others: 159.39 5 (1969Wo02), 159.2 5 (1967Ko01), 160 5 (1956Li38), 158 2 (1956Gr12), 167 2 (1955Ni15), 157 7 (1955Ly34), 185 7 (1953Ch16). $\alpha(\text{K})_{\text{exp}}$: anomalous with theory which gives $\alpha(\text{K})=0.00560 \text{ } 12$. K/L+ from visual inspection of photographic spectrometer (1953Co44). $\alpha/\alpha(\text{K})=1.12$ (1968Ba33). $\alpha(\text{K})_{\text{exp}}$: from 1986Re12 , $\text{K}/(\text{L}+\text{M}+\text{N}) \approx 10$ (1953Co44).

[†] From the Adopted Gammas.

[‡] From $\alpha(\text{K})_{\text{exp}}=0.00406 \text{ } 21$ ([1986Re12](#)) and $\text{K}/\text{L}+\approx 10$ ([1953Co44](#)). Other: 0.0036 9 from $\text{I}(\text{ce})/\text{I}\beta(439)$ ([1953Ch16](#));

[Additional information 2](#).

@ Absolute intensity per 100 decays.

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Decay Scheme

Intensities: $I_{(\gamma+ce)}$ per 100 parent decays

